Mid RC: Joint Distributions

The main question is:

> When several random variables are observed together, how do we describe their joint behavior, extract information about each variable, measure dependence, update predictions after observing information, and transform random variables into new ones?

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0. Roadmap

Part I. Joint Distributions

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1. Random Vectors and Joint Distributions

1.1 Random vector

Let

$$X_1, \dots, X_n : \Omega \to \mathbb{R}$$

be random variables. The vector

$$X = (X_1, \dots, X_n) = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} : \Omega \to \mathbb{R}^n$$

is called a **random vector** or a **multivariate random variable**.

1.2 Discrete joint pmf

Let $X = (X_1, \dots, X_n)$ be a discrete random vector. The **joint pmf** or **joint distribution** is

$$p_{X_1,\dots,X_n}(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n).$$

For a bivariate random vector (X, Y), this becomes

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

A function $p : \mathbb{R}^n \to \mathbb{R}$ is the joint pmf of some discrete random vector if

$$p(x_1, \dots, x_n) \geq 0$$

for all $(x_1, \dots, x_n)$, and

$$\sum_{(x_1,\dots,x_n) \in \mathbb{R}^n} p(x_1, \dots, x_n) = 1.$$

For any set $D \subseteq \mathbb{R}^n$,

$$P(X \in D) = \sum_{(x_1,\dots,x_n) \in D} p_{X_1,\dots,X_n}(x_1, \dots, x_n).$$

How to read a joint pmf table

If a table gives values of $p_{X,Y}(x, y)$, then each entry is a probability:

$$\text{entry in row } x \text{ and column } y = P(X = x, Y = y).$$

For example:

$$P(X = 0, Y = 0) = p_{X,Y}(0, 0).$$

1.3 Continuous joint density

Let $X = (X_1, \dots, X_n)$ be a continuous random vector. A function

$$f_{X_1,\dots,X_n} : \mathbb{R}^n \to \mathbb{R}$$

is called the **joint density** of X if

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n) \geq 0$$

for all $(x_1, \dots, x_n)$,

$$\int_{\mathbb{R}^n} f_{X_1,\dots,X_n}(x_1, \dots, x_n)\, dx_1 \cdots dx_n = 1,$$

and for any region $D \subseteq \mathbb{R}^n$,

$$P((X_1, \ldots, X_n) \in D) = \int_D f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n.$$

For two continuous random variables,

$$P((X,Y) \in D) = \iint_D f_{X,Y}(x,y)\, dxdy.$$

Important warning

For continuous random variables,

$$P(X = x, Y = y) = 0.$$

The density $f_{X,Y}(x,y)$ is not itself a probability. Probabilities are obtained by integrating the density over a region.

1.4 Joint CDF

The joint CDF of $(X_1, \ldots, X_n)$ is

$$F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = P(X_1 \le x_1, \ldots, X_n \le x_n).$$

If the random vector is continuous with joint density $f_{X_1,\ldots,X_n}$, then

$$F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{X_1,\ldots,X_n}(z_1, \ldots, z_n)\, dz_n \cdots dz_1.$$

Moreover,

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)$$

almost everywhere.

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2. Marginal Distributions

The marginal distribution describes one component of a random vector. It is obtained by summing or integrating out the other variables.

2.1 Discrete marginal pmf

Let $X = (X_1, \ldots, X_n)$ be a discrete random vector with joint pmf $p_{X_1,\ldots,X_n}$. The marginal pmf of X_k is

$$p_{X_k}(t) = \sum_{(x_1,\ldots,x_{k-1},x_{k+1},\ldots,x_n) \in \mathbb{R}^{n-1}} p_{X_1,\ldots,X_n}(x_1, \ldots, x_{k-1}, t, x_{k+1}, \ldots, x_n).$$

For bivariate random variables (X,Y),

$$p_X(x) = \sum_y p_{X,Y}(x,y),$$

and

$$p_Y(y) = \sum_x p_{X,Y}(x,y).$$

Interpretation

To compute $p_X(x)$, fix $X = x$, and add over all possible values of Y:

$$P(X = x) = \sum_y P(X = x, Y = y).$$

In a joint pmf table:

- row sums usually give $p_X(x)$,
- column sums usually give $p_Y(y)$.

2.2 Continuous marginal density

Let (X, Y) have joint density $f_{X,Y}$. The marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy,$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx.$$

More generally, for $X = (X_1, \ldots, X_n)$, the marginal density of X_k is obtained by integrating out all variables except x_k.

Technique: always determine the support first

The formal integral may be written from $-\infty$ to ∞, but in actual computation the limits are determined by the support of the joint density.

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3. Conditional Distributions

A conditional distribution describes the distribution of one variable after another variable is fixed or observed.

3.1 Discrete conditional pmf

Let (X, Y) be a discrete random vector with joint pmf $p_{X,Y}$. If $p_Y(y) > 0$, then the conditional pmf of X given $Y = y$ is

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}.$$

Equivalently,

$$p_{X|Y}(x|y) = P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}.$$

Interpretation

Fix the value $Y = y$. Then look only at the part of the table corresponding to that value of y. Divide every entry by the marginal probability $p_Y(y)$. This renormalizes the probabilities so that they sum to 1.

Indeed,

$$\sum_x p_{X|Y}(x|y) = 1.$$

3.2 Continuous conditional density

Let (X, Y) have joint density $f_{X,Y}$ and marginal densities f_X, f_Y. If $f_Y(y) > 0$, then

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

Similarly, if $f_X(x) > 0$, then

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

For fixed y, $f_{X|Y}(x|y)$ is a density as a function of x:

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)\,dx = 1.$$

Important warning

The support may change after conditioning. Always determine the conditional support.

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4. Independence

Independence means that knowing one random variable gives no information about the other.

4.1 Independence by CDF

Random variables $X_1, \dots, X_n$ are **mutually independent** if, for all $x_1, \dots, x_n \in \mathbb{R}$,

$$P(X_1 \le x_1, \dots, X_n \le x_n) = P(X_1 \le x_1) \cdots P(X_n \le x_n).$$

Equivalently,

$$F_{X_1,\dots,X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n).$$

This condition can also be stated for intervals:

$$P(a_1 < X_1 \le b_1, \dots, a_n < X_n \le b_n) = \prod_{k=1}^{n} P(a_k < X_k \le b_k).$$

4.2 Discrete independence

Discrete random variables $X_1, \dots, X_n$ are independent if and only if, for all $x_1, \dots, x_n$,

$$p_{X_1,\dots,X_n}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n).$$

For (X, Y),

$$p_{X,Y}(x, y) = p_X(x) p_Y(y)$$

for all x, y.

How to check independence in a table

To prove independence, every table entry must satisfy

$$p_{X,Y}(x, y) = p_X(x) p_Y(y).$$

To disprove independence, one counterexample is enough:

$$p_{X,Y}(x_0, y_0) \neq p_X(x_0) p_Y(y_0).$$

4.3 Continuous independence

Continuous random variables $X_1, \dots, X_n$ are independent if and only if

$$f_{X_1,\dots,X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$$

almost everywhere.

For two variables,

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

almost everywhere.

4.4 Example: rectangle support gives independence in a uniform case

Using the previous rectangle example,

$$f_{X,Y}(x, y) = \frac{1}{240}, \quad 8.5 \leq x \leq 10.5, \quad 120 \leq y \leq 240.$$

We found

$$f_X(x) = \frac{1}{2}, \qquad f_Y(y) = \frac{2}{240}.$$

Hence

$$f_X(x) f_Y(y) = \frac{1}{2} \cdot \frac{2}{240} = \frac{1}{240} = f_{X,Y}(x, y).$$

Therefore X and Y are independent.

4.5 Example: triangular support usually indicates dependence

Suppose

$$f_{X,Y}(x,y) = \frac{c}{x}, \qquad 27 \le y \le x \le 33.$$

The support is triangular, not rectangular. Once $Y = y$ is fixed, X must satisfy

$$y \le x \le 33.$$

Thus the possible values of X depend on the value of Y. This is strong evidence that X and Y are not independent.

4.6 Functions of independent random variables

If $X_1, \ldots, X_n$ are independent and $g_k : \mathbb{R} \to \mathbb{R}$, then

$$g_1(X_1), \ldots, g_n(X_n)$$

are also independent.

In particular,

$$E[g_1(X_1) \cdots g_n(X_n)] = E[g_1(X_1)] \cdots E[g_n(X_n)]$$

when the expectations exist.

Part II. Expectation and Dependence

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5. Expectation of Random Vectors and Functions

5.1 Expectation of a random vector

Let

$$X = (X_1, \ldots, X_n) : \Omega \to \mathbb{R}^n.$$

The expectation of X is the vector

$$E(X) = \begin{bmatrix} E(X_1) \\ \vdots \\ E(X_n) \end{bmatrix} \in \mathbb{R}^n.$$

5.2 Discrete case

If $X = (X_1, \ldots, X_n)$ is discrete with joint pmf $p_{X_1,\ldots,X_n}$, then

$$E(X_k) = \sum_{x_k} x_k p_{X_k}(x_k) = \sum_{(x_1,\ldots,x_n)} x_k p_{X_1,\ldots,X_n}(x_1, \ldots, x_n),$$

provided

$$\sum_{x_k} |x_k| p_{X_k}(x_k) < \infty.$$

If $\phi : \mathbb{R}^n \to \mathbb{R}$, then

$$E[\phi(X)] = \sum_{(x_1,\ldots,x_n)} \phi(x_1, \ldots, x_n) p_{X_1,\ldots,X_n}(x_1, \ldots, x_n),$$

provided the absolute convergence condition holds:

$$\sum_{(x_1,\ldots,x_n)} |\phi(x_1, \ldots, x_n)| p_{X_1,\ldots,X_n}(x_1, \ldots, x_n) < \infty.$$

For (X, Y), important special cases are

$$E(X) = \sum_x \sum_y x p_{X,Y}(x, y),$$

$$E(Y) = \sum_x \sum_y y p_{X,Y}(x, y),$$

$$E(XY) = \sum_x \sum_y xy p_{X,Y}(x, y),$$

and

$$E(X + Y) = \sum_x \sum_y (x + y) p_{X,Y}(x, y).$$

5.3 Continuous case

If $X = (X_1, \ldots, X_n)$ is continuous with joint density $f_{X_1,\ldots,X_n}$, then

$$E(X_k) = \int_{-\infty}^{\infty} x_k f_{X_k}(x_k)\, dx_k = \int_{\mathbb{R}^n} x_k f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n.$$

If $\phi : \mathbb{R}^n \to \mathbb{R}$, then

$$E[\phi(X)] = \int_{\mathbb{R}^n} \phi(x_1, \ldots, x_n) f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n,$$

provided

$$\int_{\mathbb{R}^n} |\phi(x_1, \ldots, x_n)| f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n < \infty.$$

For (X, Y),

$$E(X) = \int \int x f_{X,Y}(x, y)\, dx dy,$$

$$E(Y) = \int\int y f_{X,Y}(x, y)\, dx dy,$$

and

$$E(XY) = \int\int xy f_{X,Y}(x, y)\, dx dy.$$

5.4 Linearity of expectation

For random variables with existing expectations,

$$E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y).$$

This does **not** require independence.

However,

$$E(XY) = E(X)E(Y)$$

is not true in general. It is true if X and Y are independent and the expectations exist.

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6. Covariance, Covariance Matrix, and Correlation

6.1 Covariance

Let X, Y be random variables with means

$$\mu_X = E(X), \qquad \mu_Y = E(Y).$$

The covariance between X and Y is

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

The useful computational formula is

$$\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

Interpretation

- If X tends to be large when Y is large, covariance tends to be positive.
- If X tends to be large when Y is small, covariance tends to be negative.
- If covariance is zero, there is no linear association, but there may still be nonlinear dependence.

6.2 Properties of covariance

For constants $\alpha, \beta \in \mathbb{R}$,

$$\mathrm{Cov}(\alpha X + \beta Y, Z) = \alpha\,\mathrm{Cov}(X, Z) + \beta\,\mathrm{Cov}(Y, Z).$$

Also,

$$\operatorname{Cov}(X,Y)=\operatorname{Cov}(Y,X),$$

and

$$\operatorname{Cov}(X,X)=\operatorname{Var}(X)\ge 0.$$

If X is almost surely constant, then

$$\operatorname{Cov}(X,X)=0.$$

If $\operatorname{Var}(X)$ and $\operatorname{Var}(Y)$ are finite, then by Cauchy-Schwarz,

$$|\operatorname{Cov}(X,Y)|\le\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}.$$

If X and Y are independent, then

$$\operatorname{Cov}(X,Y)=0.$$

The converse is false in general:

$$\operatorname{Cov}(X,Y)=0\not\Rightarrow X,Y\text{ are independent.}$$

6.3 Example: uncorrelated but not independent

It is possible that

$$\operatorname{Cov}(X,Y)=0$$

but X and Y are not independent.

A typical situation is when Y determines X nonlinearly. For example, if $X=Y^2$, then X is completely determined by Y, so they cannot be independent in general. However, due to symmetry, covariance may still be zero.

Therefore covariance only detects linear dependence, not all dependence.

6.4 Covariance matrix

Let

$$X=(X_1,\dots,X_n):\Omega\to\mathbb{R}^n.$$

The covariance matrix of X is the $n\times n$ matrix

$$C_X=(c_{ij}),\qquad c_{ij}=\operatorname{Cov}(X_i,X_j).$$

That is,

$$C_X=\begin{bmatrix}\operatorname{Var}(X_1) & \operatorname{Cov}(X_1,X_2) & \cdots & \operatorname{Cov}(X_1,X_n)\\ \operatorname{Cov}(X_2,X_1) & \operatorname{Var}(X_2) & \cdots & \operatorname{Cov}(X_2,X_n)\\ \vdots & \vdots & \ddots & \vdots\\ \operatorname{Cov}(X_n,X_1) & \operatorname{Cov}(X_n,X_2) & \cdots & \operatorname{Var}(X_n)\end{bmatrix}.$$

Important properties:

$$C_X = C_X^T,$$

and C_X is positive semidefinite, meaning

$$v^T C_X v \geq 0$$

for all $v \in \mathbb{R}^n$.

If $A \in M_{m\times n}(\mathbb{R})$, then

$$C_{AX} = AC_X A^T.$$

6.5 Correlation coefficient

Let

$$\sigma_X = \sqrt{\operatorname{Var}(X)}, \qquad \sigma_Y = \sqrt{\operatorname{Var}(Y)}.$$

The correlation coefficient between X and Y is

$$\rho_{XY} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

Properties:

$$|\rho_{XY}| \leq 1.$$

Moreover,

$$|\rho_{XY}| = 1$$

if and only if there exist constants $\beta_0, \beta_1 \in \mathbb{R}$, with $\beta_1 \neq 0$, such that

$$Y = \beta_0 + \beta_1 X$$

almost surely.

Interpretation:

- $\rho_{XY} = 1$: perfect positive linear relationship.
- $\rho_{XY} = -1$: perfect negative linear relationship.
- $\rho_{XY} = 0$: no linear correlation, but nonlinear dependence may still exist.

Part III. Conditional Expectation

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7. Conditional Expectation, Regression, and Prediction

Conditional expectation is the expected value of a random variable after conditioning on another random variable.

7.1 Discrete conditional expectation

Let (X, Y) be a discrete random vector. For fixed y, the conditional expectation of X given $Y = y$ is

$$E(X \mid Y = y) = \sum_x x p_{X|Y}(x|y).$$

This is a **number**, because y is fixed.

Define

$$g(y) = E(X \mid Y = y).$$

Then

$$E(X \mid Y) = g(Y).$$

This is a **random variable**, because Y is random.

Important distinction

$$E(X \mid Y = y)$$

is a number.

$$E(X \mid Y)$$

is a random variable.

This distinction is one of the most important points in this topic.

7.2 Continuous conditional expectation

If (X, Y) has joint density $f_{X,Y}$, then

$$E(X \mid Y = y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx.$$

Again, for fixed y, this is a number.

If

$$g(y) = E(X \mid Y = y),$$

then

$$E(X \mid Y) = g(Y).$$

7.3 Example: conditional expectation on triangular support

From the triangular support example,

$$f_{X|Y}(x|y) = \frac{1}{x\ln(33/y)}, \qquad y \le x \le 33.$$

Then

$$E(X \mid Y = y) = \int_y^{33} x \cdot \frac{1}{x\ln(33/y)}\, dx.$$

Simplify the integrand:

$$x \cdot \frac{1}{x\ln(33/y)} = \frac{1}{\ln(33/y)}.$$

Therefore

$$E(X \mid Y = y) = \int_y^{33} \frac{1}{\ln(33/y)}\, dx = \frac{33 - y}{\ln(33/y)}.$$

7.4 Properties of conditional expectation

Let X, Y, Z be random variables and let $\alpha, \beta \in \mathbb{R}$. Assume all expectations exist.

7.4.1 Constants

If α is a constant, then

$$E(\alpha \mid Y) = \alpha.$$

7.4.2 Linearity

$$E(\alpha X + \beta Z \mid Y) = \alpha E(X \mid Y) + \beta E(Z \mid Y).$$

7.4.3 Positivity

If

$$X \ge 0 \quad \text{a.s.},$$

then

$$E(X \mid Y) \ge 0 \quad \text{a.s.}$$

7.4.4 Independence

If X and Y are independent, then

$$E(X \mid Y) = E(X).$$

More generally, if X is independent of Y and Z, then

$$E(XZ \mid Y) = E(X)E(Z \mid Y)$$

when the expression is meaningful.

7.4.5 Law of total expectation

$$E(E(X \mid Y)) = E(X).$$

Interpretation:

> First average X within each condition $Y = y$, then average those conditional means over the distribution of Y. The result is the overall average of X.

In the discrete case,

$$E(E(X \mid Y)) = \sum_y E(X \mid Y = y) p_Y(y) = E(X).$$

7.4.6 Taking out what is known

If $g(Y)$ is a function of Y, then

$$E(Xg(Y) \mid Y) = g(Y)E(X \mid Y).$$

This is because once Y is known, $g(Y)$ is known.

7.4.7 Conditioning on more information

The notes include the identity

$$E(X \mid Y, g(Y)) = E(X \mid Y).$$

The reason is that $g(Y)$ gives no additional information once Y is already known.

A related tower property is

$$E(E(X \mid Y, Z) \mid Y) = E(X \mid Y).$$

7.5 Independence and conditional expectation

If the random vectors

$$(X_1, \dots, X_k) \quad \text{and} \quad (Y_1, \dots, Y_m)$$

are independent, then for any suitable function f,

$$E[f(X_1, \dots, X_k) \mid Y_1, \dots, Y_m] = E[f(X_1, \dots, X_k)] \quad \text{a.s.}$$

This says: once two groups of variables are independent, conditioning on one group does not change the expectation of a function of the other group.

Also, for suitable functions f and g,

$$E[f(X_1, \dots, X_k) g(Y_1, \dots, Y_m)] = E[f(X_1, \dots, X_k)] E[g(Y_1, \dots, Y_m)].$$

7.6 Independence from a pair

If

$$X \perp (Y, Z),$$

then for suitable f,

$$E[f(X) \mid Y, Z] = E[f(X)] \quad \text{a.s.}$$

and for suitable g,

$$E[f(X)g(Y, Z)] = E[f(X)]E[g(Y, Z)].$$

A useful consequence is

$$E(XZ \mid Y) = E(X)E(Z \mid Y).$$

7.7 Conditional variance and law of total variance

7.7.1 Conditional variance

For fixed $Y = y$, define

$$\mathrm{Var}(X \mid Y = y) = E[(X - E(X \mid Y = y))^2 \mid Y = y].$$

Equivalently,

$$\mathrm{Var}(X \mid Y = y) = E(X^2 \mid Y = y) - [E(X \mid Y = y)]^2.$$

Let

$$h(y) = \mathrm{Var}(X \mid Y = y).$$

Then

$$\mathrm{Var}(X \mid Y) = h(Y),$$

which is a random variable.

Thus

$$\mathrm{Var}(X \mid Y) = E(X^2 \mid Y) - [E(X \mid Y)]^2.$$

7.7.2 Law of total variance

If $\mathrm{Var}(X) < \infty$, then

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E(X \mid Y)).$$

Interpretation:

$$\text{total variation} = \text{average within-condition variation} + \text{variation of conditional means.}$$

That is:

- $E[\mathrm{Var}(X \mid Y)]$ measures the average remaining variation of X after Y is known.

- $\mathrm{Var}(E(X \mid Y))$ measures how much the conditional mean changes as Y changes.

7.8 Curve of regression

The **curve of regression of X on Y** is the graph of

$$E(X \mid Y = y)$$

as a function of y.

The **curve of regression of Y on X** is the graph of

$$E(Y \mid X = x)$$

as a function of x.

7.9 Conditional Expectation as the Best Predictor

Suppose the goal is to estimate Y using a function of X. That is, choose a function h and use

$$h(X)$$

as a predictor of Y.

The mean squared error is

$$E[(Y - h(X))^2].$$

The best predictor in the mean squared error sense is

$$E(Y \mid X).$$

More precisely,

$$E(Y \mid X) = \arg\min_h E[(Y - h(X))^2].$$

Part IV. Gaussian Random Variables

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8. Gaussian Random Variables

This section summarizes the main facts about Gaussian random variables used in the later lectures.

8.1 Multivariate Gaussian random vector

An n-dimensional Gaussian random vector is written as

$$X \sim \mathrm{Normal}(\mu, \Sigma),$$

where

$$\mu \in \mathbb{R}^n$$

is the mean vector, and

$$\Sigma \in M_{n\times n}(\mathbb{R})$$

is the covariance matrix.

The covariance matrix must be positive semidefinite:

$$\Sigma \ge 0, \qquad v^T \Sigma v \ge 0 \quad \forall v \in \mathbb{R}^n.$$

If Σ is positive definite, then the density exists and is

$$f_X(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right].$$

Here:

- $x \in \mathbb{R}^n$ is a vector of ordinary real values;
- μ determines the center of the distribution;
- Σ determines the spread and correlation structure.

8.2 Bivariate normal density

For a bivariate normal random vector (X, Y), write

$$\mu = \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{bmatrix}.$$

Then ρ is the correlation coefficient:

$$\rho = \rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

The joint density is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}.$$

The standard bivariate normal case is

$$\mu_X = \mu_Y = 0, \qquad \sigma_X = \sigma_Y = 1.$$

Then

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right].$$

8.3 Gaussian independence

For general random variables,

$$\mathrm{Cov}(X, Y) = 0$$

does **not** imply independence.

However, for jointly Gaussian random variables,

$$X, Y \text{ jointly Gaussian} \quad \Longrightarrow \quad X \perp Y \iff \mathrm{Cov}(X, Y) = 0.$$

Equivalently, in the bivariate normal case,

$$X \perp Y \iff \rho = 0.$$

This is a special property of the Gaussian distribution.

8.4 Closure properties of Gaussian random vectors

Gaussian random vectors are closed under linear transformations.

If

$$Z \sim \mathrm{Normal}(\mu_Z, \Sigma_Z),$$

then

$$AZ + b \sim \mathrm{Normal}(A\mu_Z + b, A\Sigma_Z A^T).$$

This is useful because many new Gaussian variables are built as linear combinations of old ones.

A key characterization is:

$$X = (X_1, \dots, X_n)^T \text{ is Gaussian}$$

if and only if every linear combination

$$a_1 X_1 + \cdots + a_n X_n$$

is a one-dimensional Gaussian random variable.

8.5 Block notation for covariance matrices

Suppose

$$X \in \mathbb{R}^m, \qquad Y \in \mathbb{R}^n.$$

Then the joint vector

$$\begin{bmatrix} X \\ Y \end{bmatrix}$$

has a covariance matrix that can be written in block form:

$$\Sigma = \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}.$$

The blocks mean:

$$\Sigma_{XX} = \mathrm{Cov}(X, X), \qquad \Sigma_{YY} = \mathrm{Cov}(Y, Y),$$

and

$$\Sigma_{XY} = \mathrm{Cov}(X, Y), \qquad \Sigma_{YX} = \mathrm{Cov}(Y, X) = \Sigma_{XY}^T.$$

Thus Σ_{XX} and Σ_{YY} describe the variation inside X and inside Y, while Σ_{XY} and Σ_{YX} describe the cross-covariance between X and Y.

8.6 Marginalization of Gaussian random vectors

Suppose

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathrm{Normal}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right).$$

Then the marginal distribution of X is

$$X \sim \mathrm{Normal}(\mu_X, \Sigma_{XX}).$$

Similarly,

$$Y \sim \mathrm{Normal}(\mu_Y, \Sigma_{YY}).$$

8.7 Conditioning of Gaussian random vectors

With the same block notation,

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathrm{Normal}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right),$$

the conditional distribution of X given $Y = y$ is also Gaussian:

$$X \mid (Y = y) \sim \mathrm{Normal}\left(\mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y), \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}\right).$$

The conditional mean is linear in y:

$$E(X \mid Y = y) = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y).$$

The conditional covariance is

$$\mathrm{Cov}(X \mid Y = y) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}.$$

This matrix is called the Schur complement of Σ_{YY} in Σ.

8.8 First-quadrant probability for standard bivariate normal

If

$$(X, Y) \sim N\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\right),$$

then

$$P(X > 0, Y > 0) = \frac{1}{4} + \frac{1}{2\pi}\arcsin(\rho).$$

This formula explains how positive correlation moves more probability mass into the same-sign quadrants, and negative correlation moves probability mass into opposite-sign quadrants.

For example, if $\rho = 0$, then X and Y are independent, and

$$P(X > 0, Y > 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

If $\rho > 0$, then

$$P(X > 0, Y > 0) > \frac{1}{4}.$$

If $\rho < 0$, then

$$P(X > 0, Y > 0) < \frac{1}{4}.$$

Part V. Transformation of Variables

9. Transformation of Variables

The goal of transformation of variables is to find the density of a new random vector defined from an old one.

9.1 One-dimensional change of variables

Suppose X has density f_X, and

$$Y = g(X)$$

where g is one-to-one and differentiable. Let

$$x = g^{-1}(y).$$

Then

$$f_Y(y) = f_X(g^{-1}(y)) \left|\frac{d}{dy} g^{-1}(y)\right|.$$

Equivalently,

$$f_Y(y) = f_X(x(y)) \left| \frac{dx}{dy} \right|.$$

The absolute derivative is the one-dimensional Jacobian factor.

9.2 Multivariate change of variables

Let

$$X = (X_1, \ldots, X_n)$$

have joint density f_X. Suppose

$$Y = g(X),$$

where

$$g : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$$

is C^1 and injective. Let g^{-1} be the inverse map.

Then the density of Y is

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left|\det J_{g^{-1}}(y)\right|, & y \in g(U), \\ 0, & \text{otherwise.} \end{cases}$$

Here $J_{g^{-1}}(y)$ is the Jacobian matrix of the inverse map:

$$J_{g^{-1}}(y) = \left[\frac{\partial x_i}{\partial y_j} \right]_{i,j=1}^{n}.$$

The determinant factor accounts for how volume changes under the transformation.

9.3 Linear transformation example

Suppose X and Y are independent uniform random variables on $(0, 2)$ and $(0, 3)$, respectively. Then

$$f_{X,Y}(x, y) = \frac{1}{6}, \qquad 0 < x < 2, \quad 0 < y < 3.$$

Define

$$U = X - Y, \qquad V = X + Y.$$

The transformation is

$$(u, v) = g(x, y) = (x - y, x + y).$$

Solving for x, y,

$$x = \frac{u + v}{2}, \qquad y = \frac{v - u}{2}.$$

The inverse Jacobian determinant is

$$\left|\det J_{g^{-1}}(u, v)\right| = \frac{1}{2}.$$

Therefore

$$f_{U,V}(u,v) = f_{X,Y}\left(\frac{u+v}{2}, \frac{v-u}{2}\right) \cdot \frac{1}{2} = \frac{1}{12}$$

on the transformed support.

The support is determined by translating

$$0 < x < 2, \qquad 0 < y < 3$$

into conditions on u, v:

$$0 < \frac{u+v}{2} < 2, \qquad 0 < \frac{v-u}{2} < 3.$$

Equivalently,

$$0 < u+v < 4, \qquad 0 < v-u < 6.$$

9.4 Product transformation

Suppose (X, Y) has joint density $f_{X,Y}$. We want the density of

$$U = XY.$$

A common trick is to introduce an auxiliary variable

$$V = X.$$

Then

$$(u,v) = g(x,y) = (xy, x).$$

The inverse is

$$x = v, \qquad y = \frac{u}{v}, \qquad v \neq 0.$$

The absolute inverse Jacobian factor is

$$\left|\det J_{g^{-1}}(u,v)\right| = \frac{1}{|v|}.$$

Thus

$$f_{U,V}(u,v) = f_{X,Y}\left(v, \frac{u}{v}\right)\frac{1}{|v|}.$$

Finally, integrate out v:

$$f_U(u) = \int_{-\infty}^{\infty} f_{X,Y}\left(v, \frac{u}{v}\right)\frac{1}{|v|}\, dv.$$

If X and Y are independent, then

$$f_{X,Y}(x,y) = f_X(x) f_Y(y),$$

so

$$
f_U(u) = \int_{-\infty}^{\infty} f_X(v) f_Y\left(\frac{u}{v}\right) \frac{1}{|v|}\, dv.
$$

9.5 Polar coordinate example: $X^2 + Y^2$

Let

$$
X, Y \sim N(0,1)
$$

be independent. Define

$$
Z = X^2 + Y^2.
$$

Since

$$
f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2},
$$

we compute the CDF of Z:

$$
F_Z(z) = P(Z \le z) = P(X^2 + Y^2 \le z).
$$

Use polar coordinates:

$$
x = r\cos\theta, \qquad y = r\sin\theta, \qquad dxdy = r\, drd\theta.
$$

For $z > 0$, the region $x^2 + y^2 \le z$ becomes

$$
0 \le r \le \sqrt{z}, \qquad 0 \le \theta \le 2\pi.
$$

Thus

$$
F_Z(z) = \int_0^{2\pi} \int_0^{\sqrt{z}} \frac{1}{2\pi} e^{-r^2/2} r\, drd\theta = 1 - e^{-z/2}.
$$

Therefore

$$
f_Z(z) = F_Z'(z) = \frac{1}{2} e^{-z/2}, \qquad z > 0.
$$

So

$$
Z = X^2 + Y^2 \sim \chi_2^2.
$$

9.6 Sum of independent continuous random variables: convolution

Let X and Y be independent continuous random variables with densities f_X and f_Y. Define

$$
Z = X + Y.
$$

Then

$$
f_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y)\, dy.
$$

This is called the convolution of f_X and f_Y:

$$f_Z = f_X * f_Y.$$

A useful way to remember the formula is:

$$X + Y = z \quad \Longleftrightarrow \quad X = z - y.$$

Then integrate over all possible values of $Y = y$.

Part VI. Classical Derivations and Applications

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10. Classical Distribution Derivations

10.1 Sum of independent Poisson random variables

Let

$$X \sim \text{Poisson}(\mu), \qquad Y \sim \text{Poisson}(\lambda),$$

and assume X and Y are independent.

For $n = 0, 1, 2, \ldots$,

$$P(X + Y = n) = \sum_{k=0}^{n} P(X = k, Y = n - k).$$

By independence,

$$P(X + Y = n) = \sum_{k=0}^{n} P(X = k) P(Y = n - k).$$

Substitute the Poisson pmfs:

$$P(X + Y = n) = \sum_{k=0}^{n} e^{-\mu} \frac{\mu^k}{k!} e^{-\lambda} \frac{\lambda^{n-k}}{(n-k)!}.$$

Factor out common terms:

$$P(X + Y = n) = e^{-(\mu+\lambda)} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} \mu^k \lambda^{n-k}.$$

By the binomial theorem,

$$\sum_{k=0}^{n} \binom{n}{k} \mu^k \lambda^{n-k} = (\mu + \lambda)^n.$$

Therefore

$$P(X + Y = n) = e^{-(\mu+\lambda)} \frac{(\mu + \lambda)^n}{n!},$$

so

$$X + Y \sim \text{Poisson}(\mu + \lambda).$$

10.2 Conditional normal example

Let

$$Z_1 \sim \mathrm{Normal}(\mu_1, \sigma_1^2), \qquad Z_2 \sim \mathrm{Normal}(\mu_2, \sigma_2^2),$$

where Z_1, Z_2 are independent and $\sigma_1, \sigma_2 > 0$.

We want the distribution of

$$Z_1 \mid (Z_1 + Z_2 = z).$$

Consider the vector

$$\begin{bmatrix} Z_1 \\ Z_1 + Z_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}.$$

Because Gaussian random vectors are closed under linear transformation,

$$\begin{bmatrix} Z_1 \\ Z_1 + Z_2 \end{bmatrix}$$

is bivariate normal.

Its mean vector is

$$\begin{bmatrix} \mu_1 \\ \mu_1 + \mu_2 \end{bmatrix},$$

and its covariance matrix is

$$\begin{bmatrix} \sigma_1^2 & \sigma_1^2 \\ \sigma_1^2 & \sigma_1^2 + \sigma_2^2 \end{bmatrix}.$$

Using the Gaussian conditioning formula,

$$Z_1 \mid (Z_1 + Z_2 = z) \sim \mathrm{Normal}(\widetilde{\mu}, \widetilde{\sigma}^2),$$

where

$$\widetilde{\mu} = \mu_1 + \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}(z - \mu_1 - \mu_2),$$

and

$$\widetilde{\sigma}^2 = \sigma_1^2 - \frac{\sigma_1^4}{\sigma_1^2 + \sigma_2^2} = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

Interpretation: once the sum is fixed, the uncertainty of Z_1 is reduced.

10.3 Sum of iid exponential random variables

Let

$$W_1, W_2, \ldots$$

be iid exponential random variables with parameter $\lambda > 0$:

$$f_W(w) = \lambda e^{-\lambda w}, \qquad w > 0.$$

Define

$$S_n = W_1 + \cdots + W_n.$$

Then

$$S_n$$

has an Erlang distribution, also called a Gamma distribution with integer shape n:

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \qquad t > 0.$$

This formula can be derived by convolution, by using the joint density of arrival times, or by relating S_n to a Poisson process.

If N_t is a Poisson process with rate λ, then

$$\{S_n \leq t\} = \{N_t \geq n\}.$$

Differentiating the CDF gives the same density.

---

11. Buffon's Needle Problem

Buffon's needle problem is a geometric probability problem.

A needle of length ℓ is dropped on a plane ruled with parallel lines separated by distance d. We ask for the probability that the needle crosses one of the lines.

11.1 Random variables

Let

$$X = \text{distance from the center of the needle to the closest line},$$

and

$$\Theta = \text{angle between the needle and the parallel lines}.$$

By symmetry,

$$0 < X < \frac{d}{2}, \qquad 0 < \Theta < \frac{\pi}{2}.$$

Assume

$$X \sim \text{Uniform}\left(0, \frac{d}{2}\right), \qquad \Theta \sim \text{Uniform}\left(0, \frac{\pi}{2}\right),$$

and X and Θ are independent.

Therefore the joint density is

$$f_{X,\Theta}(x, \theta) = \frac{4}{\pi d}, \qquad 0 < x < \frac{d}{2}, \quad 0 < \theta < \frac{\pi}{2}.$$

11.2 Crossing condition

The needle crosses a line if the vertical half-length of the needle reaches the closest line.

The vertical half-length is

$$\frac{\ell}{2} \sin \theta.$$

Therefore crossing occurs if and only if

$$X \leq \frac{\ell}{2} \sin \Theta.$$

11.3 Short needle: $\ell \leq d$

If $\ell \leq d$, then

$$\frac{\ell}{2} \sin \theta \leq \frac{d}{2}$$

for all $0 \leq \theta \leq \pi/2$. Hence the crossing probability is

$$p = \int_0^{\pi/2} \int_0^{(\ell/2) \sin \theta} \frac{4}{\pi d} \, dx \, d\theta.$$

Compute the inner integral:

$$p = \int_0^{\pi/2} \frac{4}{\pi d} \cdot \frac{\ell}{2} \sin \theta \, d\theta = \frac{2\ell}{\pi d} \int_0^{\pi/2} \sin \theta \, d\theta.$$

Since

$$\int_0^{\pi/2} \sin \theta \, d\theta = 1,$$

we get

$$\boxed{p = \frac{2\ell}{\pi d}}.$$

11.4 Long needle: $\ell > d$

If $\ell > d$, then for some angles the vertical half-length exceeds $d/2$, so the needle crosses automatically for those angles.

The crossing probability can be written as

$$p = \frac{2}{\pi} \int_0^{\pi/2} \min\left(1, \frac{\ell}{d} \sin\theta\right) d\theta.$$

Let

$$\theta_0 = \arcsin\left(\frac{d}{\ell}\right).$$

Then

$$p = \frac{2}{\pi} \left[\int_0^{\theta_0} \frac{\ell}{d} \sin\theta \, d\theta + \int_{\theta_0}^{\pi/2} 1 \, d\theta \right].$$

Thus

$$p = \frac{2}{\pi} \left[\frac{\ell}{d} \left(1 - \sqrt{1 - \left(\frac{d}{\ell}\right)^2} \right) + \arccos\left(\frac{d}{\ell}\right) \right].$$

11.5 Linearity of expectation viewpoint

A very elegant solution for the short needle result uses linearity of expectation.

Let N_ℓ be the number of crossings made by a needle of length ℓ. For short needles, the crossing probability is

$$P(N_\ell = 1) = E(N_\ell).$$

Expected crossings are additive with respect to length. If a needle of length 1 has expected crossings p_1, then a needle of length ℓ has expected crossings

$$E(N_\ell) = \ell p_1.$$

A circular needle or curve argument gives the calibration

$$p_1 = \frac{2}{\pi d}.$$

Therefore

$$E(N_\ell) = \frac{2\ell}{\pi d},$$

and for $\ell \le d$, this equals the crossing probability:

$$P(\text{crossing}) = \frac{2\ell}{\pi d}.$$

---

12. Summary of Main Formulas

This section collects the main formulas from all four lectures.

12.1 Discrete joint pmf

$$p_{X,Y}(x,y) = P(X = x, Y = y).$$

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y).$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}, \qquad p_Y(y) > 0.$$

$$E[\phi(X,Y)] = \sum_x \sum_y \phi(x,y) p_{X,Y}(x,y).$$

12.2 Continuous joint density

$$\int\int f_{X,Y}(x,y)\, dxdy = 1.$$

$$P((X,Y) \in D) = \iint_D f_{X,Y}(x,y)\, dxdy.$$

$$f_X(x) = \int f_{X,Y}(x,y)\, dy, \qquad f_Y(y) = \int f_{X,Y}(x,y)\, dx.$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \qquad f_Y(y) > 0.$$

$$E[\phi(X,Y)] = \int\int \phi(x,y) f_{X,Y}(x,y)\, dxdy.$$

12.3 Joint CDF and density

$$F_{X_1,\dots,X_n}(x_1,\dots,x_n) = P(X_1 \le x_1, \dots, X_n \le x_n).$$

$$f_{X_1,\dots,X_n}(x_1,\dots,x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1,\dots,X_n}(x_1,\dots,x_n) \quad \text{a.e.}$$

12.4 Independence

Discrete:

$$p_{X,Y}(x,y) = p_X(x) p_Y(y).$$

Continuous:

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad \text{a.e.}$$

CDF version:

$$F_{X,Y}(x,y) = F_X(x) F_Y(y).$$

Functions of independent variables are independent. If $X \perp Y$, then

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)].$$

If $X \perp Y$, then

$$E(X \mid Y) = E(X).$$

12.5 Covariance and correlation

$$\operatorname{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))].$$

$$\operatorname{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

$$\operatorname{Cov}(\alpha X + \beta Y, Z) = \alpha \operatorname{Cov}(X, Z) + \beta \operatorname{Cov}(Y, Z).$$

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}.$$

$$|\rho_{XY}| \leq 1.$$

$$|\rho_{XY}| = 1 \iff Y = \beta_0 + \beta_1 X \quad \text{a.s. for some } \beta_1 \neq 0.$$

12.6 Covariance matrix

For

$$X = (X_1, \ldots, X_n)^T,$$

the covariance matrix is

$$C_X = (c_{ij}), \qquad c_{ij} = \operatorname{Cov}(X_i, X_j).$$

$$C_X = C_X^T, \qquad v^T C_X v \geq 0.$$

If $A \in M_{m \times n}(\mathbb{R})$, then

$$C_{AX} = A C_X A^T.$$

12.7 Conditional expectation and variance

Discrete:

$$E(X \mid Y = y) = \sum_x x p_{X|Y}(x|y).$$

Continuous:

$$E(X \mid Y = y) = \int x f_{X|Y}(x|y)\, dx.$$

As a random variable,

$$E(X \mid Y) = g(Y), \qquad g(y) = E(X \mid Y = y).$$

Linearity:

$$E(\alpha X + \beta Z \mid Y) = \alpha E(X \mid Y) + \beta E(Z \mid Y).$$

Taking out what is known:

$$E(Xg(Y) \mid Y) = g(Y)E(X \mid Y).$$

Law of total expectation:

$$E(E(X \mid Y)) = E(X).$$

Tower property:

$$E(E(X \mid Y, Z) \mid Y) = E(X \mid Y).$$

Conditional variance:

$$\mathrm{Var}(X \mid Y) = E(X^2 \mid Y) - [E(X \mid Y)]^2.$$

Law of total variance:

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E(X \mid Y)).$$

Best predictor:

$$E(Y \mid X) = \arg\min_h E[(Y - h(X))^2].$$

12.8 Regression curves

Curve of regression of X on Y:

$$y \mapsto E(X \mid Y = y).$$

Curve of regression of Y on X:

$$x \mapsto E(Y \mid X = x).$$

12.9 Multivariate Gaussian formulas

$$X \sim \mathrm{Normal}(\mu, \Sigma)$$

has density, if $\Sigma > 0$,

$$f_X(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right].$$

Linear transformation:

$$AZ + b \sim \mathrm{Normal}(A\mu_Z + b, A\Sigma_Z A^T).$$

Gaussian conditioning:

$$X \mid (Y = y) \sim \mathrm{Normal}\left(\mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y), \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}\right).$$

Jointly Gaussian independence:

$$X \perp Y \iff \mathrm{Cov}(X, Y) = 0.$$

Standard bivariate normal first-quadrant probability:

$$P(X > 0, Y > 0) = \frac{1}{4} + \frac{1}{2\pi}\arcsin(\rho).$$

12.10 Transformation of variables

One-dimensional:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Multivariate:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \det J_{g^{-1}}(y) \right|.$$

Product:

$$U = XY \quad \Longrightarrow \quad f_U(u) = \int_{-\infty}^{\infty} f_{X,Y}\left(v, \frac{u}{v}\right) \frac{1}{|v|}\, dv.$$

Sum of independent continuous variables:

$$Z = X + Y \quad \Longrightarrow \quad f_Z(z) = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy.$$

This is convolution:

$$f_Z = f_X * f_Y.$$

Polar coordinate Jacobian:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad dxdy = r\, drd\theta.$$

12.11 Classical distribution results

Poisson sum:

$$X \sim \mathrm{Poisson}(\mu), \quad Y \sim \mathrm{Poisson}(\lambda), \quad X \perp Y$$

implies

$$X + Y \sim \mathrm{Poisson}(\mu + \lambda).$$

Sum of iid exponential variables:

$$W_i \sim \mathrm{Exponential}(\lambda), \quad S_n = W_1 + \cdots + W_n$$

implies

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \qquad t > 0.$$

Standard normal square sum:

$$X, Y \sim N(0,1), \quad X \perp Y \quad \Longrightarrow \quad X^2 + Y^2 \sim \chi_2^2.$$

12.12 Buffon's needle

Let ℓ be the needle length and d be the distance between parallel lines.

$$X \sim \text{Uniform}\left(0, \frac{d}{2}\right), \qquad \Theta \sim \text{Uniform}\left(0, \frac{\pi}{2}\right).$$

Crossing condition:

$$X \leq \frac{\ell}{2} \sin \Theta.$$

Short needle, $\ell \leq d$:

$$P(\text{crossing}) = \frac{2\ell}{\pi d}.$$

General formula for $\ell > d$:

$$P(\text{crossing}) = \frac{2}{\pi} \left[\frac{\ell}{d} \left(1 - \sqrt{1 - \left(\frac{d}{\ell}\right)^2} \right) + \arccos\left(\frac{d}{\ell}\right) \right].$$

h4 ex 4.4

X is always positive.

$$F_Y(y) = P(Y<y) = P\left(\tfrac{1}{2}mX^2<y\right) = P\left(X<\sqrt{\tfrac{2y}{m}}\right) = P_X\left(\sqrt{\tfrac{2y}{m}}\right) \Rightarrow$$ take derivative

$$f_Y(y) = f_X\left(\sqrt{\tfrac{2y}{m}}\right)\cdot\frac{1}{\sqrt{2my}} = c\frac{\sqrt{2}}{m^{\frac{3}{2}}}\,y^{\frac{1}{2}}\,e^{-2\beta y/m}, \quad y>0.$$

h4 ex 4.6

To show that $X \sim \text{Poisson}(p\lambda)$, we need to show that

for $k=0,1,2\ldots$, we get $P(X=k) = e^{-p\lambda}\frac{(p\lambda)^k}{k!}$

$$P(X=k) = \sum_{n=0}^{\infty} P(X=k\mid N=n)\cdot P(N=n)$$

↓ when $n<k$, impossible to choose k from n.

$$= \sum_{n=k}^{\infty} P(X=k\mid N=n)\cdot P(N=n)$$

↓ unfold def

$$= \sum_{n=k}^{\infty}\binom{n}{k}p^k(1-p)^{n-k}\cdot e^{-\lambda}\frac{\lambda^n}{n!}$$

↓ unfold $\binom{n}{k}$, simplify

$$= \sum_{n=k}^{\infty}\frac{1}{k!(n-k)!}p^k(1-p)^{n-k}e^{-\lambda}\lambda^n$$

↓ rearrange so that exp serious appears

$$= \frac{e^{-\lambda}(p\lambda)^k}{k!}\sum_{n=k}^{\infty}\frac{((1-p)\lambda)^{n-k}}{(n-k)!}$$

↓ substitute $j:=n-k$

$$= \frac{e^{-\lambda}(p\lambda)^k}{k!}\sum_{j=0}^{\infty}\frac{((1-p)\lambda)^{j}}{j!}$$

↓ fold the series into $e^{(1-p)\lambda}$ and simplify.

$$= e^{-p\lambda}\frac{(p\lambda)^k}{k!}$$

Thus, $X \sim \text{Poisson}(p\lambda)$.

* Optional: Given a standard bivariate normal pair (X,Y), i.e. $(X,Y)\sim N\left(\begin{pmatrix}0\\0\end{pmatrix},\begin{pmatrix}1&\rho\\\rho&1\end{pmatrix}\right)$, calculate $P(X>0,Y>0)$.

Given that $f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(-\frac{x^2-2\rho xy+y^2}{2(1-\rho^2)}\right)$

1° Convert into marginal x conditional:

rewrite to $f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(-\frac{(y-\rho x)^2+(1-\rho^2)x^2}{2(1-\rho^2)}\right)$

$$= \left(\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)\right)\left(\frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}}\exp\left(-\frac{(y-\rho x)^2}{2(1-\rho^2)}\right)\right)$$

$f_X(x)$ $\qquad$ $f_{Y|X}(y|x) \Rightarrow Y|X=x \sim N(\rho x, 1-\rho^2)$

2° Convert to integration over one variable.

notation: Φ standard N CDF

$\phi := \Phi'$.

$$P(X>0, Y>0) = \int_0^\infty P(Y>0 \mid X=x)\, f_X(x)\, dx$$

As $Y \mid X=x \sim N(\rho x, 1-\rho^2)$, $P(Y>0 \mid X=x) = P\left(\frac{Y-\rho x}{\sqrt{1-\rho^2}} > \frac{0-\rho x}{\sqrt{1-\rho^2}}\right) = P\left(Z > -\frac{\rho x}{\sqrt{1-\rho^2}}\right)$

$$= \Phi\left(\frac{\rho x}{\sqrt{1-\rho^2}}\right).$$

Thus, $P(X>0, Y>0) = \int_0^\infty \phi(x)\, \Phi\left(\frac{\rho x}{\sqrt{1-\rho^2}}\right) dx$

Set $a := \frac{\rho}{\sqrt{1-\rho^2}}$. denote $I(a) := \int_0^\infty \phi(x)\, \Phi(ax)\, dx$.

To calculate $I(a)$, we calculate:

$$I'(a) = \frac{d}{da}\int_0^\infty \phi(x)\Phi(ax)\,dx = \int_0^\infty \phi(x)\frac{\partial}{\partial a}\Phi(ax)\,dx$$

$$= \int_0^\infty \phi(x)\cdot x\cdot \phi(ax)\,dx = \int_0^\infty x\cdot\frac{1}{2\pi}e^{-\frac{(1+a^2)x^2}{2}}\,dx$$

$$= \int_0^\infty e^{-u}\frac{du}{1+a^2} \quad \swarrow \text{ set } u = \frac{(1+a^2)x^2}{2}$$

$$= \frac{1}{2\pi(1+a^2)}$$

Thus, $I(a) = \frac{1}{2\pi}\arctan(a) + \frac{1}{4}$ (constant part can be determined by calculating $I(0)$).

Substitute $a = \frac{\rho}{\sqrt{1-\rho^2}}$, we get $P(X>0, Y>0) = \frac{1}{4} + \frac{1}{2\pi}\arctan\left(\frac{\rho}{\sqrt{1-\rho^2}}\right) = \frac{1}{4} + \frac{1}{2\pi}\arcsin(\rho)$.

h5 ex 5.1

The first thing we need is to standardize + completing the square

Set $u := \frac{x-\mu_X}{\sigma_X}$, $v = \frac{y-\mu_Y}{\sigma_Y}$, the quadratic part becomes

$$u^2 - 2\rho uv + v^2 = (u-\rho v)^2 + (1-\rho^2)u^2.$$

Thus, $f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left(-\frac{u^2}{2}\right)\exp\left(-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right)$.

(i) $f_X = \int_{-\infty}^\infty f_{X,Y}(x,y)\,dy$

$v = \frac{y-\mu_Y}{\sigma_Y}$

$dy = \sigma_Y\, dv$

$$= \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left(-\frac{u^2}{2}\right)\int_{-\infty}^\infty \exp\left(-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right)dv$$

Gaussian integral: $\int_{-\infty}^\infty \exp\left(-\frac{(v-a)^2}{2s^2}\right)dv = \sqrt{2\pi}s$

$$= \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}\exp\left(-\frac{u^2}{2}\right)\sqrt{2\pi}\sqrt{1-\rho^2}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_X}\exp\left(-\frac{u^2}{2}\right) = \frac{1}{\sqrt{2\pi}\sigma_X}\exp\left(-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2\right)$$

Thus, $X \sim N(\mu_X, \sigma_X^2)$.

(ii) By def, we need to show that $Cov(X,Y) = \rho \sigma_X \sigma_Y$.

We first standardize $U := \frac{X-\mu_X}{\sigma_X}$, $V := \frac{Y-\mu_Y}{\sigma_Y}$.

As $Cov(X,Y) = E((X-\mu_X)(Y-\mu_Y)) = \sigma_X\sigma_Y E(UV)$, the problem is reduced to prove $E(UV) = \rho$.

Instead of directly integrating, consider $E(UV) = E(E(UV|U)) = E(UE(V|U))$

As the density for (U,V) is $f_{U,V}(u,v) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{u^2-2\rho uv+v^2}{2(1-\rho^2)}\right)$

$$= \underbrace{\left(\frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}\right)}_{U} \cdot \underbrace{\left(\frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left(-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right)\right)}_{V|U=u}$$

Thus $U \sim N(0,1)$, $V|U=u \sim N(\rho u, 1-\rho^2)$

Thus, $E(V|U) = \rho U \Rightarrow E(UE(V|U)) = \rho E(U^2) = \rho(1-0) = \rho \Rightarrow E(UV) = \rho$.

Therefore, $Corr(X,Y) = \rho$.

(iii) $\Rightarrow$: $0 = Cov(X,Y) = \rho\sigma_X\sigma_Y$. As $\sigma_X > 0$, $\sigma_Y > 0$, $\rho = 0$

$\Leftarrow$: $\rho = 0 \Rightarrow f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left(-\frac{1}{2}(u^2+v^2)\right) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left(-\frac{1}{2}u^2\right) \cdot \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left(-\frac{1}{2}v^2\right)$

$= f_X(x) f_Y(y) \Rightarrow X \perp Y$

(iv) $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right)$

$$= \frac{1}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2}\left(\frac{y-m}{\sigma_Y\sqrt{1-\rho^2}}\right)^2\right)$$

where $m := \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)$

Thus, $Y|X=x \sim N(\underbrace{\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)}_{E}, \sigma_Y^2(1-\rho^2))$.

h5 ex 5.2

Inspecting the structure of the formula:

When y, z is fixed, $\exp\left(-\frac{1}{2}\left(\frac{x-y}{z}\right)^2\right)$ is Gaussian kernel for x.

When z is fixed, e^{-zy} is exponential.

Thus, the structure behind is $\begin{cases} X|Y,Z \sim N(Y, Z^2) \\ Y|Z \sim \text{Exponential}(Z). \end{cases}$

(i) To prove: 1° non neg ✓ 2° $\int = 1$

2° $\int_1^2 \int_0^\infty \int_{-\infty}^\infty f_{X,Y,Z}(x,y,z)\,dx\,dy\,dz = \int_1^2\int_0^\infty \frac{3z^2}{7\sqrt{2\pi}} e^{-zy} \int_{-\infty}^\infty \exp\left(-\frac{1}{2}\left(\frac{x-y}{z}\right)^2\right) dx\,dy\,dz$

$= \int_1^2\int_0^\infty \frac{3z^2}{7\sqrt{2\pi}} e^{-zy} \cdot \sqrt{2\pi}\, z\, dy\,dz = \int_1^2 \frac{3z^3}{7} \int_0^\infty e^{-zy} dy\,dz = \int_1^2 \frac{3z^3}{7} \cdot \frac{1}{z}\, dz = 1$.

(ii) $f_{Y,Z}(y,z) = \int_{-\infty}^{\infty} f_{X,Y,Z}\, dx = \frac{3z^3}{7} e^{-zy}$, $y \geq 0$, $1 \leq z \leq 2$.

$$f_{X|Y,Z}(x|y,z) = \frac{f_{X,Y,Z}}{f_{Y,Z}} = \frac{1}{\sqrt{2\pi}\, z} \exp\left(-\frac{1}{2}\left(\frac{x-y}{z}\right)^2\right) \Rightarrow X|Y=y, Z=z \sim N(y, z^2).$$

(iii) $f_Z(z) = \int_0^{\infty} f_{Y,Z}(y,z)\, dy = \frac{3z^2}{7}$, $1 \leq z \leq 2$

$$f_{Y|Z}(y|z) = \frac{f_{Y,Z}}{f_Z} = z \cdot e^{-zy}, \quad y \geq 0 \Rightarrow Y|Z=z \sim \text{Exponential}(z)$$

$$f_{X,Y|Z}(x,y|z) = \frac{f_{X,Y,Z}}{f_Z} = \frac{1}{\sqrt{2\pi}} e^{-zy} \exp\left(-\frac{1}{2}\left(\frac{x-y}{z}\right)^2\right), \quad y \geq 0$$

(iv) In this ex, we will utilize tower property : $E(X) = E(E(X|Y))$.

As we already have $X|Y,Z \sim N(Y, Z^2)$, $E(X|Y,Z) = Y$.

$Y|Z \sim \text{Exponential}(Z)$, $E(Y|Z) = \frac{1}{Z}$, $E(Y^2|Z) = \frac{2}{Z^2}$.

Thus, every 3-dim integration can be reduced to simply on Z.

1° $E(X) = E(E(X|Y,Z)) = E(Y) = E(E(Y|Z)) = E(\frac{1}{Z}) = \int_1^2 \frac{1}{z} \cdot \frac{3z^2}{7}\, dz = \frac{9}{14}$

2° $E(XZ) = E(E(XZ|Y,Z)) = E(ZE(X|Y,Z)) = E(ZY) = E(E(ZY|Z)) = E(ZE(Y|Z))$

$= E(Z \cdot \frac{1}{Z}) = 1$.

3° $E(XY) = E(E(XY|Y,Z)) = E(YE(X|Y,Z)) = E(Y^2) = E(E(Y^2|Z)) = E(\frac{2}{Z^2})$

$= \int_1^2 \frac{2}{z^2} \cdot \frac{3z^2}{7}\, dz = \frac{6}{7}$

4° $E(YZ) = E(E(YZ|Z)) = E(ZE(Y|Z)) = E(Z \cdot \frac{1}{Z}) = 1$

5° $E(XYZ) = E(E(XYZ|Y,Z)) = E(YZE(X|Y,Z)) = E(Y^2Z) = E(E(ZY^2|Z)) = E(ZE(Y^2|Z))$

$= E(Z \cdot \frac{2}{Z^2}) = 2E(\frac{1}{Z}) = 2 \cdot \frac{9}{14} = \frac{9}{7}$.